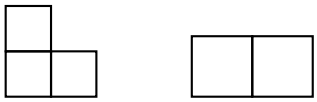
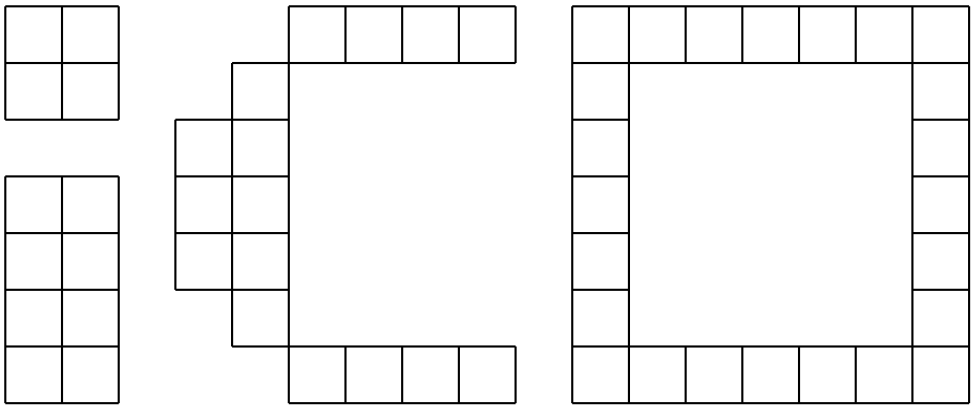
Combinatorics Olympiad
Intermediate Level
October 2025

Problems

1. The domino and L-tromino pieces are as shown below.



In how many ways can we tile the given shape shown in the figure below using some number of dominoes and L-trominoes, with no overlaps?
Note: rotations of pieces are allowed.

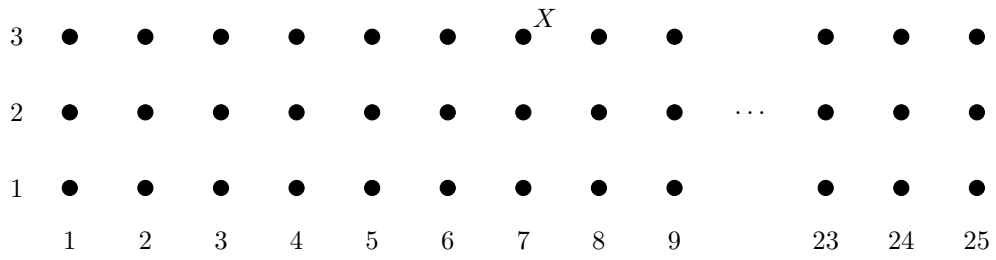


2. 117 gangsters are standing in a line. Simultaneously, each gangster shoots the nearest person to himself. (If two people are equally close — one on each side — the gangster may choose either of them.) What is the minimum possible number of gangsters who get shot?

3. Suppose n is an even number, and n people are seated around a circle. We say a number d is *interesting for* n if it is possible to divide these people into $\frac{n}{2}$ disjoint pairs such that in each pair, one person is exactly the d-th person clockwise after the other. Find the number of even integers n such that $100 \leq n \leq 200$, where every positive integer d less than n is interesting for n.

4. The houses of a village are arranged on the points of a 3×25 rectangular grid, as shown in the figure below. That is, the village has 75 houses, one at each grid point. A house located in row i and column j of the grid is labeled (i, j). For instance, the house marked by X is labeled $(3, 7)$. Traveling from house (i, j) to house (i', j') using jet takes $\sqrt{(i - i')^2 + (j - j')^2}$ minutes.

   A family living in $(1, 1)$ wants to visit all the other houses (each house exactly once), and then return to their own house. They use jet for traveling between houses. Assume that this family wants to visit the house $(2, 2)$ immediately after leaving (their house at) $(1, 1)$. In how many different ways can they complete this process while minimizing the total time?



5. Let n be a natural number. Consider the graph with vertices

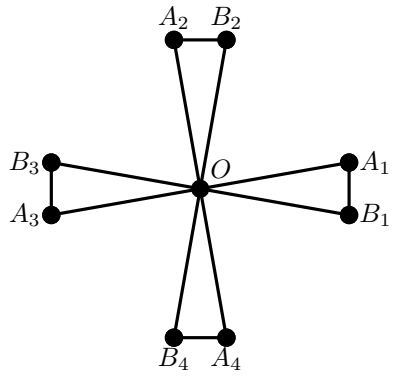
$$O, A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_n$$

   and edges

$$OA_1, OA_2, \ldots, OA_n, OB_1, OB_2, \ldots, OB_n$$

$$A_1B_1, A_2B_2, \ldots, A_nB_n.$$

   We denote this graph by S_n. For example, the structure of S_4 is illustrated below:



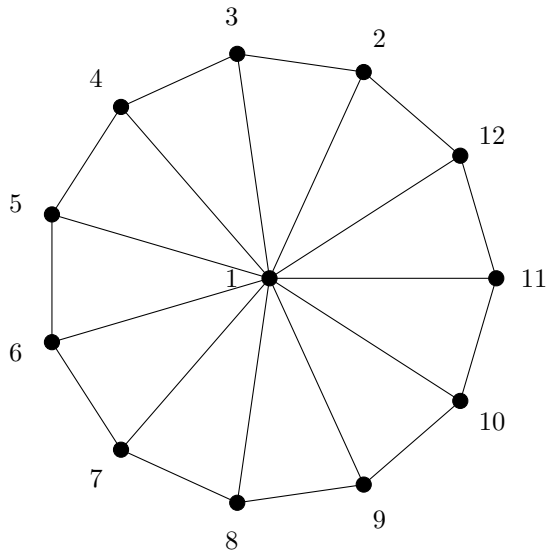
   In how many ways can we color the edges of S_{15} red and blue so that each vertex is incident to at least one blue edge and at least one red edge?

6. Dara and Sara play the following game:

   > There are 100 points labeled 1 through 100 on the number line. Initially, all points are uncolored. In each move, Dara chooses one of the uncolored points, and Sara colors it either black or white.
   >
   > After each turn, if it is possible to draw a straight line such that all white points lie on one side of the line and all black points lie on the other side of the line, Dara earns one coin. (Indeed, if all the colored points so far are of the same color, Dara earns one coin.)

   Dara wants to maximize the number of coins that he earns, and Sara wants to minimize that. Assuming both play optimally, find the number of coins that Dara has at the end of the game.

7. In how many ways can we color some of the segments in the following figure red so that each of the 12 points is connected to an odd number of red segments?



8. The education system of a country consists of exactly seven subjects. Each school must schedule these seven subjects during the seven days of the week, exactly one subject per day.

   The school employs seven teachers (one per subject). Each teacher proposes two possible days of the week on which they can teach their subject as their availabilities.

   Suppose that based on the proposed days by the teachers, the school can schedule the weekly program in x ways. What is the maximum possible value of x?

9. Suppose 2025 boys are seated around a circle. Two boys are called friends if the number of boys sitting between them is an even number smaller than 809. (For example, each boy is friend with both of his immediate neighbors.) In how many ways can we choose three boys such that any two of them are friends?

   Note: The order of selection does not matter.

10. Hamid's Grandpa has passed away, and his old suitcase has been inherited by Hamid. The lock on this suitcase is a two-digit display, initially showing 00. For this lock, there is a unique two-digit pass-code. As soon as the correct pass-code appears on the display, the lock opens.

    There are four buttons:

    - Button 1 increases the right digit by 1.
    - Button 2 decreases the right digit by 1.
    - Button 3 increases the left digit by 1.
    - Button 4 decreases the left digit by 1.

    A digit cannot be increased if it is already 9, and cannot be decreased if it is already 0.

    Hamid does **not** know the correct pass-code, but he remembers his grandfather saying that (1) neither 09 nor 90 is the lock pass-code, and (2) if the display ever shows 09 or 90, the suitcase will be locked forever.

    What is the minimum number of times that Hamid must press the buttons to ensure that he will definitely open the lock, regardless of what the correct pass-code is?

11. Consider a 2×10 table. In how many ways can the numbers 1 to 20 be placed in the cells such that:

    - All numbers are distinct.
    - For any two cells sharing a side, their corresponding numbers differ by at most 2.

12. Ali defends his castle with 20 soldiers, and Hassan attacks it with 10 soldiers. The battle has three fronts. Ali wins the war if, on each front, he deploys at least as many soldiers as Hassan does.

    Each of Ali and Hassan independent of the other, chooses one of all possible ways of deploying their soldiers, each with equal probability. (For example, Ali might choose $(0, 12, 8)$, meaning 0 soldiers on the first front, 12 on the second, and 8 on the third.)

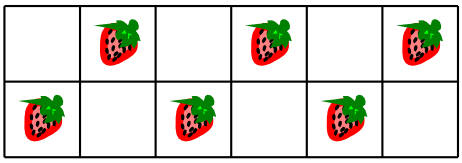
    Suppose that the probability that Ali wins the war is $\frac{p}{q}$, where p and q are co-prime positive integers. What is the value of $p + q$?

13. 10 uncolored balls with numbers 1 to 10 are arranged around a circle. In how many ways can we color each ball by red or blue so that (1) there are exactly 5 blue and 5 red balls, and (2) for each red ball, among the two balls immediately to its left and the two balls immediately to its right, there are at least two red balls in total?

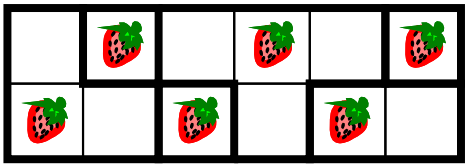
Strawberry farm

A strawberry farm is a 2×6 table such that there is a strawberry in some cells. We want to divide the table along its table lines into 6 connected regions such that each region contains exactly one strawberry. We call such partitioning a *valid partitioning*.

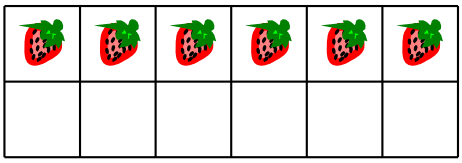
14. If the farm is filled with strawberries as in the figure below, how many valid partitionings exists?



An example of a valid partition is shown in the figure below:



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